MATH 263 Notes
Chapter 6 - Set Theory

Natalie Piltoyan

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Contents

Definitions and the Element Method of Proof **2**

Properties of Sets **5**

Definitions and the Element Method of Proof

A *set* is an unordered collection of distinct objects. Each of those objects are called *elements* of the set. To say that an element a is in a set S, we write $a \in S$. To say that a is not in the set S, we write $a \notin S$.

Say we have a set A, and we want to define a new a set that contains all elements of A that satisfy some property. To define such a set S, we use the following notation:

$$S = \{x \in A \mid P(x)\}$$

This is read as "S is the set of all elements x in the set A that satisfy $P(x)$".

Examples

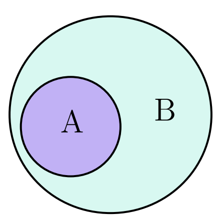
- Let $x = \frac{1}{2}$. Then $x \in \mathbb{Q}$, but $x \notin \mathbb{Z}$.
- Let $S = \{x \in \mathbb{N} \mid x \bmod 3 = 1\}$. Then $S = \{1, 4, 7, 10, 13, \ldots\}$.

Definition 1. *Let A, B be sets. We say that A is a **subset** of B, or $A \subseteq B$, if, and only if, for all $x \in A$, $x \in B$.*
*We say that A is a **proper subset** of B, written $A \subset B$, if, and only if, $A \subseteq B$ and there is at least one element x so that $x \in B$ and $x \notin A$.*

A set A is a subset of another set B if every element in A is also in B. As a universal conditional statement, this is: $A \subseteq B \iff \forall x$, if $x \in A$, then $x \in B$.

This definition allows for A and B to be the same set - if $A = B$, then every element of A will be in B. A is a proper subset of B if they are not equal, so there has to be at least one element of B that is not in A. If A is not a subset of B, we write $A \subsetneq B$.

Below is a visual description of $A \subseteq B$.



To prove that $A \subseteq B$, we start with an arbitrary element x that is in A, then show that x must also be in B.

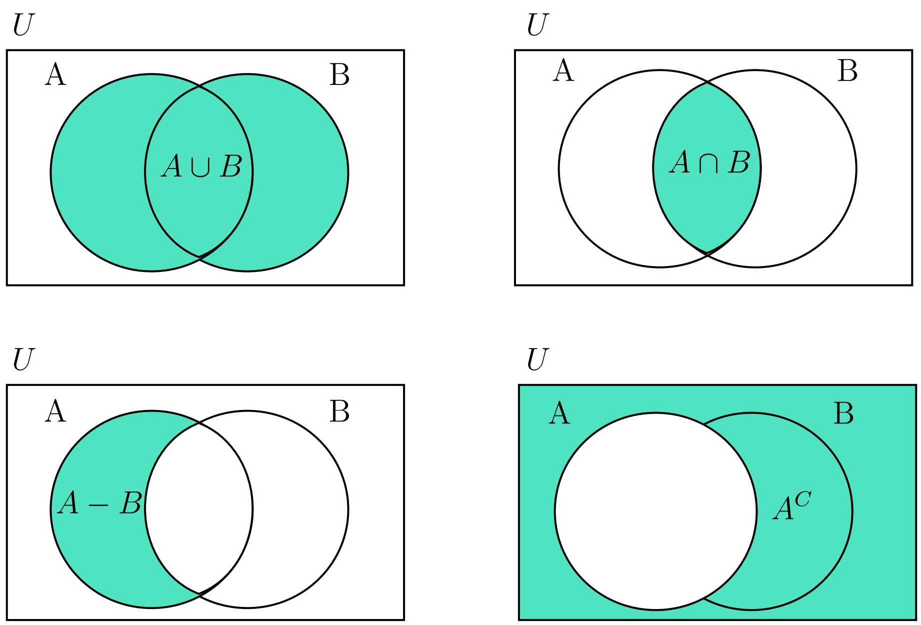
Let A, B be sets. We say that A **equals** B, written $A = B$, if and only if every element of A is in B and every element of B is in A. To show that two sets are equal, we show that $A \subseteq B$ and that $B \subseteq A$.

Below are some operations on sets.

Definition 2. *Let A, B be subsets of a universal set U.*

- *The **union** of A and B, denoted $A \cup B$, is the set of all elements that are in A or B.*
- *The **intersection** of A and B, denoted $A \cap B$, is the set of all elements that are in A and B.*
- *The **difference** of A minus B, denoted $A - B$, is the set of all elements that are in A but not in B.*
- *The **complement** of A, denoted A^C, is the set of all elements in U that are not in A.*

Below is a visual representation of these operations.



We can take the union or intersection of a collection of sets - if we have a collection of n sets $A_1, A_2, \ldots, A_n$, then

- $A_1 \cup A_2 \cup \cdots \cup A_n$ can be written as $\bigcup_{i=1}^{n} A_i$
- $A_1 \cap A_2 \cap \cdots \cap A_n$ can be written as $\bigcap_{i=1}^{n} A_i$

The *empty set* (sometimes called the *null set*) is the set that contains no elements. We denote it with $\emptyset$. By default, $\emptyset$ is a subset of every single set.

Definition 3. *Let A, B be sets. We say A and B are **disjoint** if, and only if, $A \cap B = \emptyset$. Sets $A_1, A_2, \ldots, A_n$ are **mutually disjoint** if, and only if, no two sets have any elements in common.*

This is saying that two sets are disjoint if they have no elements in common. If we have a collection of sets, they are all mutually disjoint if none of them share elements.

Definition 4. *Let A be a set. The* ***power set*** *of A, denoted $\mathcal{P}(A)$, is the set of all subsets of A.*

Be careful with this one - each element of $\mathcal{P}(A)$ should be a set, **not** a regular element. The number of elements in the power set is always a power of 2: if A has n elements, then the power set has 2^n elements.

Example

- Let $A = \{x, y\}$. Write out the power set of A.
  Solution $\mathcal{P}(A) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}$.
- Let $A = \{1, 2, 3\}$. Write out the power set.
  Solution $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$.

Properties of Sets